Exercise sheet 4

PRISM - Curves and Surfaces

1. Recall the torsion $\tau(t)$ that we defined for a curve parametrized by θ at the point $\theta(t)$.
   (a) Can you see why the definition of a torsion does not depend on the parametrization? In other words, it remains the same after reparametrizing.
   (b) Derive a formula to obtain the torsion from even a non-unit speed parametrization.
   (c) Show that the for a plane curve the torsion is always 0.
   (d) Show that if the torsion is always 0, then the curve must lie on a plane. (*Hint:* look at the point $\theta(t)$ from the perspective of the coordinates given by $\mathbf{T}(t)$, $\mathbf{N}(t)$, and $\mathbf{B}(t)$).
2. If I give you two functions $\tau : (a,b) \to \mathbb{R}^3$ and $\kappa : (a,b) \to \mathbb{R}^3$, will you always be find a unit speed parametrization $\theta : (a,b) \to \mathbb{R}^3$ so that its torsion will be τ and its curvature will be κ? (*Hint:* If you know the rate of change of a function at each point and the initial value, can you recover the function?)
3. For these questions, recall the multivariable chain rule that we discussed in class: If $F : \mathbb{R}^2 \to \mathbb{R}$ is a two variable function and $\theta : \mathbb{R} \to \mathbb{R}^3$ is a parametrization of a curve in $\mathbb{R}^2$ with coordinate functions, say x and y so that $\theta(t) = (x(t), y(t))$, then we can compute the derivative of the composition $F(\theta(t))$ (which can be interpreted as the rate of change of F along the curve θ), in terms of F and θ as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}F(\theta(t)) = x'(t)\frac{\partial f}{\partial x} + y'(t)\frac{\partial f}{\partial y}$$

   where $\frac{\partial f}{\partial x}$ is the derivative obtained by keeping y fixed and treating the function as a function only in x. Similarly, we can define $\frac{\partial f}{\partial x}$.
   (a) Use the chain rule to derive a formula of "implicit differentiation" (which you must have learned in school) in the following way: If $F : \mathbb{R}^2 \to \mathbb{R}$ is a function and $y = f(x)$ for some function $f : \mathbb{R} \to \mathbb{R}$ so that $F(x,y) = 0$, then show that $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{\partial F/\partial x}{\partial F/\partial y}$.
   (b) Show that if θ_2 is another parametrization of the same curve but with the same velocity vector at the point $\theta(t)$ then $\frac{\mathrm{d}}{\mathrm{d}t}F(\theta_2(t)) = \frac{\mathrm{d}}{\mathrm{d}t}F(\theta(t))$